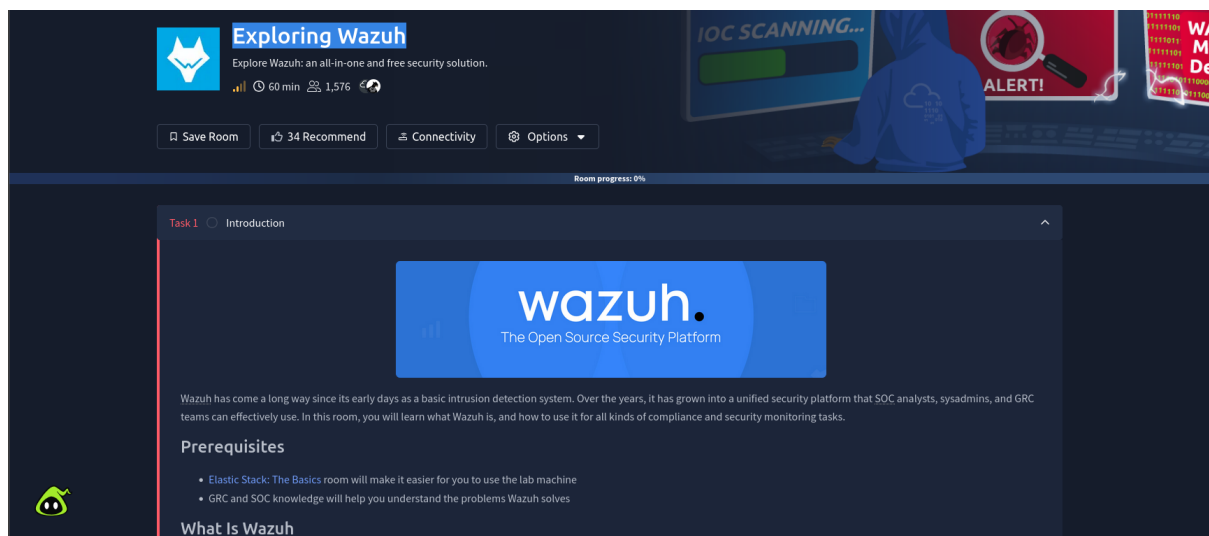


Exploring Wazuh



Wazuh originally started as a host-based intrusion detection system (HIDS), but it has evolved into a broader unified security platform. Today, Wazuh brings together capabilities such as:

- Endpoint Detection and Response (EDR)** — monitors endpoints for suspicious behavior.
- Security Information and Event Management (SIEM)** — collects, analyzes, and correlates security events.
- Vulnerability Detection** — identifies vulnerable software and packages on monitored systems.
- File Integrity Monitoring (FIM)** — detects unauthorized changes to important files.
- Security Configuration Assessment (SCA)** — checks systems against security configuration standards.
- Cloud Security Monitoring** — monitors activity and configuration in cloud environments.
- Compliance Monitoring** — helps organizations assess systems against regulatory/security frameworks.
- Dashboards and Visualization** — turns large amounts of security data into understandable alerts, graphs, and reports.

Task 2: Launching the Wazuh Server

Task 2 ☐ Launch Wazuh Server

Wazuh Architecture

Founded in 2015, Wazuh is used across organizations of all sizes - from small businesses to large enterprises and government institutions. Wazuh operates on a manager and agent model. Simply put, there is one Wazuh server (**manager**) that stores and processes the data, and many hosts (**agents**) that send data to the manager. Let's look at this model in the diagram below:



We can see logs from four Agents being sent to the Wazuh server

In more technical terms, the Wazuh server is a complex deployment consisting of multiple components: an indexer built on top of OpenSearch (a fork of Elasticsearch), a custom-built orchestrator (manager), Filebeat that sends data from the manager to the indexer, and a GUI built on top of OpenSearch dashboards. In this room, all components are installed on a single VM.

Deploy the Wazuh management server attached to this task and wait **4-5 minutes** before visiting the Wazuh server. Once it has started, log in using the following credentials:

- URL: `https://10-49-129-33.reverse-proxy.cell-prod-ap-south-1b.vm.tryhackme.com:8443`
- Username: `thmuser`
- Password: `TryHackMe!`

Your virtual environment has been set up ✓
All machine details can be found at the top of the page.

**Lab machine** ⓘ
Status: ● On

Open machine in browser ↗

Once you log in, you should see the Overview page with two agents installed. Please note that the agents will be in the disconnected state in your lab; this is expected.

W. Overview

API default a

AGENTS SUMMARY

Active (2)

Disconnected (0)

LAST 24 HOURS ALERTS

Critical severity
0
Rule level 15 or higher

High severity
1
Rule level 12 to 14

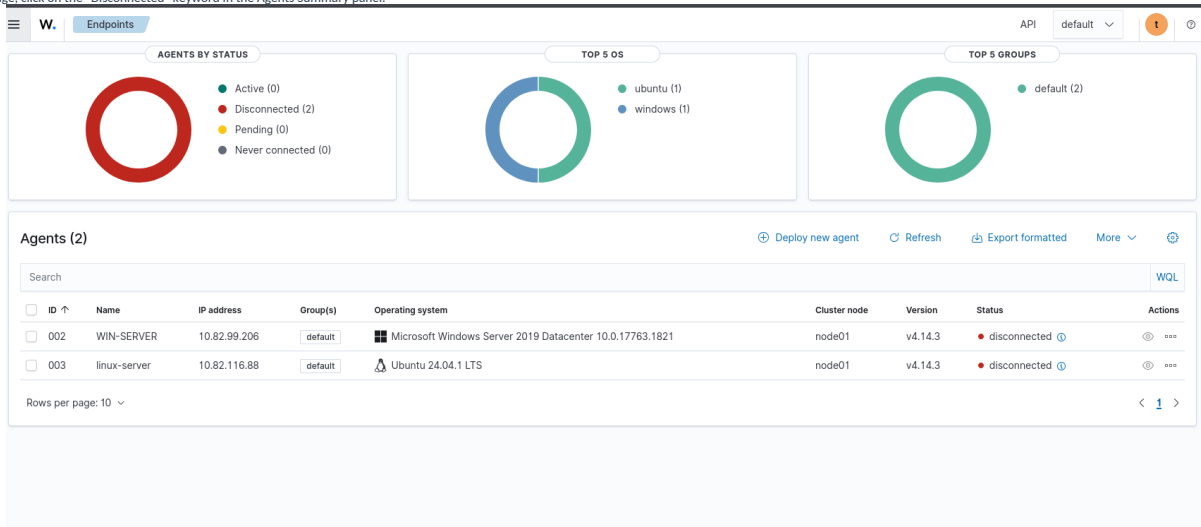
Medium severity
996
Rule level 7 to 11

Low severity
4,685
Rule level 0 to 6

Answer: No answer needed.

Task 3: Installing Wazuh Agents

Agents monitor the events that take place on the device, such as authentication and OS activities; they also audit the device for misconfigurations and much more. Let's analyze the two existing agents: one Linux and one Windows device. From the Overview page, click on the "Disconnected" keyword in the Agents Summary panel:



Agent Groups

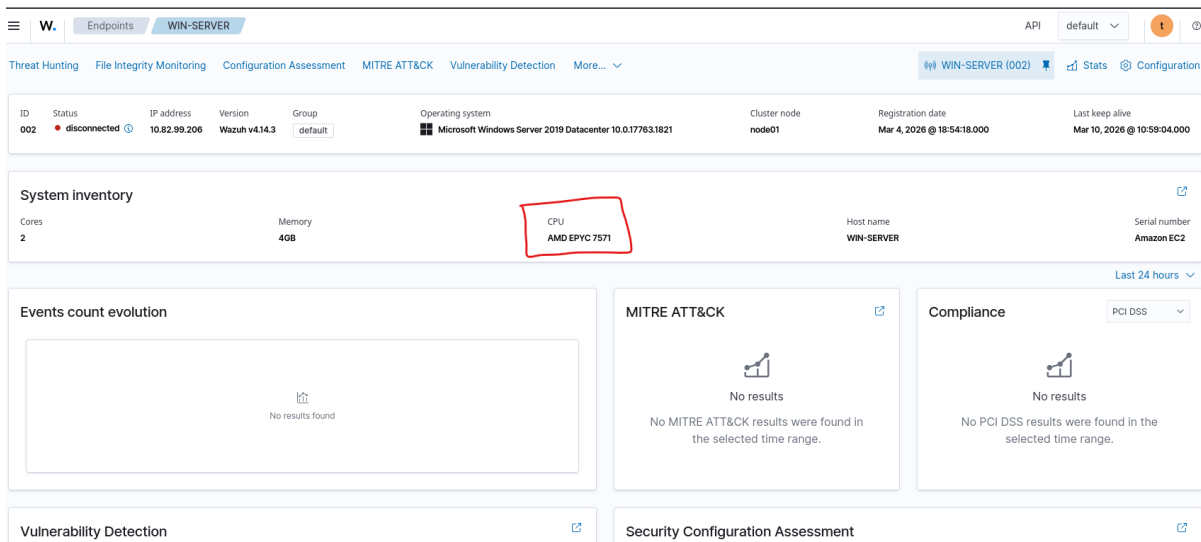
If you were to deploy an agent, you would need to specify the IP address of your Wazuh server, the agent name (typically the same as the hostname), and the group. Agents can be organized into groups, each of which can contain specific audit policies and logging settings. For example, you could have a group named **CORP** for corporate servers and desktops, and another group for personal laptops.

Question: What is the status of the agents managed by this Wazuh?

Answer: Disconnected.

Question: Click on the Windows agent. What is the CPU field value?

Answer: AMD EPYC 7571



Task 4: Configuration Assessment

Once the agent is installed, Wazuh runs lots of security checks and collects a ton of information about the host. One of the most useful categories for IT is the IT Hygiene checks. Select the WIN-SERVER agent, click **More**, and open the **IT Hygiene** section. Once you are there, you will see multiple tabs, such as:

The screenshot shows the Wazuh dashboard for the WIN-SERVER agent. A dropdown menu is open, showing the following options:

- Endpoint security**
 - Configuration Assessment (highlighted)
 - Malware Detection
 - File Integrity Monitoring
- Security operations**
 - IT Hygiene (highlighted)
 - PCI DSS
 - GDPR
 - HIPAA
 - NIST 800-53
 - TSC
- Threat intelligence**
 - Threat Hunting
 - Vulnerability Detection
 - MITRE ATT&CK
- Cloud security**
 - Amazon Web Services
 - Google Cloud
 - GitHub
 - Office 365
 - Microsoft Graph API

The dashboard also displays the following sections:

- System inventory:** Cores: 2, Memory: 4GB.
- Events count evolution:** No results found.
- Vulnerability Detection:** 21 Critical, 988 High.
- Top 5 Packages:**

Package	Count
Microsoft Windows Server 2019 Datacenter 10.0.17763.1821	1461
- Security Configuration Assessment:**

Policy	End scan	Passed	Failed	Not appl...	Score
CIS Microsoft Windows Server 2019 Benchmark v2.0.0	Mar 10, 2026 @ 07:34:30.000	85	262	0	24%
- Compliance:** No results.

System: Hardware, hostname, network interfaces, and OS version.
Software: Installed programs, OS patches, and even browser extensions.
Network: Listening ports (vital for exposure management).
Identity: Local users present on the device, including their privileges.

The screenshot shows the Wazuh dashboard for the WIN-SERVER agent, specifically the IT Hygiene section. The dashboard displays the following sections:

- Dashboard:** System, Software, Processes, Network, Identity, Services.
- Search:** wazuh.cluster.name: tryhackme-2404, agent.id: 002.
- Operating system families:**

Family	Count
windows	1
- Package types:** win.
- Top 5 installed packages:**

Package	Count
Microsoft Url Search Hook	3
AWS PV Drivers	1
AWS Tools for Windows	1
Amazon SSM Agent	1
Microsoft Visual C++ 2015-2019 Redistributable	1
- Top 5 running processes:**

Process	Count
svchost.exe	55
RuntimeBroker.exe	3
dllhost.exe	3
fontdrvhost.exe	3
LogonUI.exe	2
- Top 5 operating systems:**

OS	Count
Microsoft Windows Server 2019 Datacenter	1
- Top 5 host CPUs:**

CPU	Count
AMD EPYC 7571	1
- Top 5 endpoints by memory usage:**

Endpoint	Total memory	Usage
WIN-SERVER	4GB	45%
- Top 5 destination ports:** 443, 3389.
- Top 5 source ports:** 3389, 443.
- Processes start time:** 80.

Wazuh and CIS Benchmarks

Wazuh is capable of auditing and monitoring an agent's configuration against CIS benchmarks or custom checklists. As shown in the image below, when the Wazuh agent is installed on the Windows machine, it runs 347 CIS security checks (typically PowerShell commands) and verifies the results. You can use this to audit your devices for misconfigurations and prepare them for security audits. Select the WIN-SERVER agent and open its **Configuration Assessment** panel to get started:

W. Configuration Assessment WIN-SERVER API default 1

Dashboard Inventory Events WIN-SERVER (002)

CIS MICROSOFT WINDOWS SERVER 2019 BENCHMARK

Passed (85) Failed (262) Not applicable (0)

CIS Microsoft Windows Server 2019 Benchmark v2.0.0

Passed 85 Failed 262 Not applicable 0 Score 24% End scan Mar 10, 2026 @ 07:34:30.000

Checks (347) Refresh Export formatted

Search WQL

ID	Title	Target	Result
16500	Ensure 'Maximum password age' is set to '365 or fewer days, but...	Command: C:\Windows\SysNative\WindowsPowerShell\v1.0\powershell secedit /export /c /cf \$env:TEMP\secpol.cfg; Get-Content \$env:TEMP\secpol.cfg Select-String "MaximumPasswordAge"; Remove-Item \$env:TEMP\secpol.cfg	Passed
16501	Ensure 'Minimum password age' is set to '1 or more day(s)'.	Command: C:\Windows\SysNative\WindowsPowerShell\v1.0\powershell secedit /export /c /cf \$env:TEMP\secpol.cfg; Get-Content \$env:TEMP\secpol.cfg Select-String "MinimumPasswordAge"; Remove-Item \$env:TEMP\secpol.cfg	Failed
16502	Ensure 'Minimum password length' is set to '14 or more characte...	Command: C:\Windows\SysNative\WindowsPowerShell\v1.0\powershell secedit /export /c /cf \$env:TEMP\secpol.cfg; Get-Content	Failed

Policy: CIS Microsoft Windows Server 2019 Benchmark v2.0.0

Rows per page: 15 < 1 >

- Question: Open the Windows agent and navigate to its IT Hygiene tab. What custom text editor is installed there (Software > Packages)?
- Answer: Notepad++

W. IT Hygiene WIN-SERVER API default 1

Dashboard Inventory Events

Top 5 vendors

Vendor	Count
Amazon Web Services	3
Microsoft Corporation	3
Amazon Web Services Developer Relations	1
Notepad++ Team	1
Wazuh, Inc.	1

9 Unique packages

Package types

9 hits

Export Formatted Reset view 22 available fields Columns Density Sort fields Full screen

agent.name	package.vendor	package.name	package.version	package.type
WIN-SERVER	Microsoft Corporation	Microsoft Visual C++ 2015-2019 Redistribut...	14.28.29910.0	win
WIN-SERVER	Amazon Web Services	AWS PV Drivers	8.3.4	win
WIN-SERVER	Wazuh, Inc.	Wazuh Agent	4.14.3	win
WIN-SERVER	Microsoft Corporation	Microsoft Visual C++ 2019 X64 Additional Run...	14.28.29910	win
WIN-SERVER	Amazon Web Services	Amazon SSM Agent	3.0.529.0	win
WIN-SERVER	Amazon Web Services	aws-cfn-bootstrap	2.0.5	win
WIN-SERVER	Microsoft Corporation	Microsoft Visual C++ 2019 X64 Minimum Runt...	14.28.29910	win
WIN-SERVER	Amazon Web Services Developer Relations	AWS Tools for Windows	3.15.1248	win
WIN-SERVER	Notepad++ Team	Notepad++ (64-bit x64)	8.4	win

- Question: Now open the Linux agent and its Configuration Assessment tab. What is the CIS Benchmark score of the Linux agent (e.g., 65%)?
- Answer: 46%

W. Configuration Assessment linux-server API default 1

Dashboard Inventory Events linux-server (003)

CIS UBUNTU LINUX 24.04 LTS BENCHMARK V1.0.0

Passed (110) Failed (127) Not applicable (42)

CIS Ubuntu Linux 24.04 LTS Benchmark v1.0.0

Passed 110 Failed 127 Not applicable 42 Score 46% End scan Mar 10, 2026 @ 02:21:26.000

Checks (279) Refresh Export formatted

Search WQL

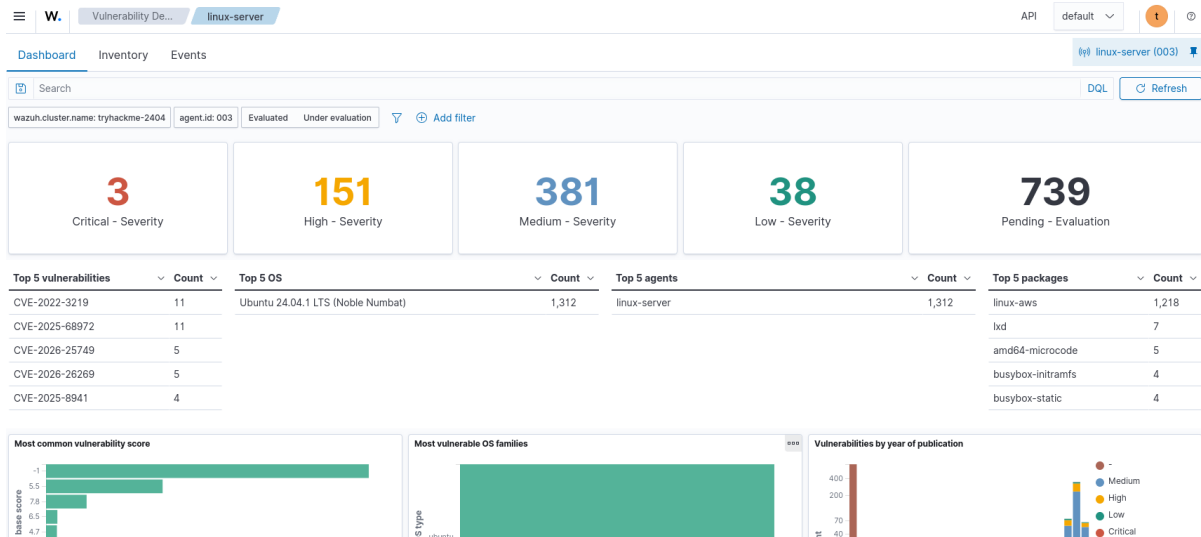
ID	Title	Target	Result
35500	Ensure mounting of cramfs filesystems is disabled.	Command: modprobe -n -v cramfs	Passed
35501	Ensure freevxfs kernel module is not available.	Command: modprobe -n -v freevxfs	Passed
35502	Ensure hfs kernel module is not available.	Command: modprobe -n -v hfs	Passed
35503	Ensure hfsplus kernel module is not available.	Command: modprobe -n -v hfsplus	Passed
35504	Ensure jfs2 kernel module is not available.	Command: modprobe -n -v jfs2	Passed
35505	Ensure overlays kernel module is not available.	Command: modprobe -n -v overlays	Passed
35506	Ensure squashfs kernel module is not available.	Command: modprobe -n -v squashfs,smmod	Passed
35507	Ensure udf kernel module is not available.	Command: modprobe -n -v udf,lsmod	Passed

Policy: CIS Ubuntu Linux 24.04 LTS Benchmark v1.0.0

Rows per page: 15 < 1 >

Task 5: Vulnerability Detection

Wazuh's Vulnerability Detection module is another powerful tool that can be used to periodically scan an agent for vulnerable applications. However, out of the box, it is arguably sensitive. Take, for example, the fresh Ubuntu server named **linux-server**. During the latest scan, Wazuh reported 1,293 vulnerabilities, 3 of which are critical (you might see even more in your lab). While the information is technically correct, only a few of the reported vulnerabilities can be exploited, and IT administrators would need to decide what to patch on their own.



Wazuh reevaluates vulnerabilities from time to time and every time a new package is installed. You can sort and filter vulnerabilities based on various factors such as timestamp, CVE number, package, or description. For example, to filter for Linux kernel vulnerabilities, you should select the **linux-server** agent, go to the **Inventory** sub-tab, and filter for **package.name IS linux-aws**.

Edit filter

Field: package.name Operator: is Value: linux-aws

☐ Create custom label?

Table Headers: agent.name, package.name, package.version, vulnerability.description, vulnerability.severity, vulnerability.id

Table Data (selected rows):

agent.name	package.name	package.version	vulnerability.description	vulnerability.severity	vulnerability.id
linux-server	snapt				CVE-2024-5138
linux-server	libde265-0			Medium	CVE-2024-38950
linux-server	libde265-0			Medium	CVE-2024-38949
linux-server	coreutils			Medium	CVE-2016-2781
linux-server	polkitd			Medium	CVE-2016-2568
linux-server	login	1:4.13+dfsg1-4ubuntu3.2	shadow-utils (aka shadow) 4.4 through...	Low	CVE-2024-56433
linux-server	libnfts-3g89t64	1:2022.10.3-1.2ubuntu3	NTFS-3G before 75dcdc2 has a use-a...	Medium	CVE-2023-52890
linux-server	libpolkit-agent-1-0	124-2ubuntu1	pkexec, when used with --user nonpri...	High	CVE-2016-2568
linux-server	busybox-static	1:1.36.1-6ubuntu3.1	In tar in BusyBox through 1.37.0, a TA...	Low	CVE-2025-46394
linux-server	busybox-static	1:1.36.1-6ubuntu3.1	A heap-buffer-overflow was discovere...	Medium	CVE-2023-42366
linux-server	busybox-static	1:1.36.1-6ubuntu3.1	BusyBox wget thru 1.3.7 accepted ra...	Medium	CVE-2025-60876
linux-server	busybox-static	1:1.36.1-6ubuntu3.1	In netstat in BusyBox through 1.37.0, l...	Low	CVE-2024-58251
linux-server	passwd	1:4.13+dfsg1-4ubuntu3.2	shadow-utils (aka shadow) 4.4 through...	Low	CVE-2024-56433
linux-server	libbpf1	1:1.3.0-2build2	Buffer Overflow vulnerability in libbpf ...	Medium	CVE-2025-29481

Question: What is the latest Notepad++ vulnerability found on the Windows agent? Provide the most recent CVE number (e.g., CVE-2026-23569).
Answer: CVE-2026-25926

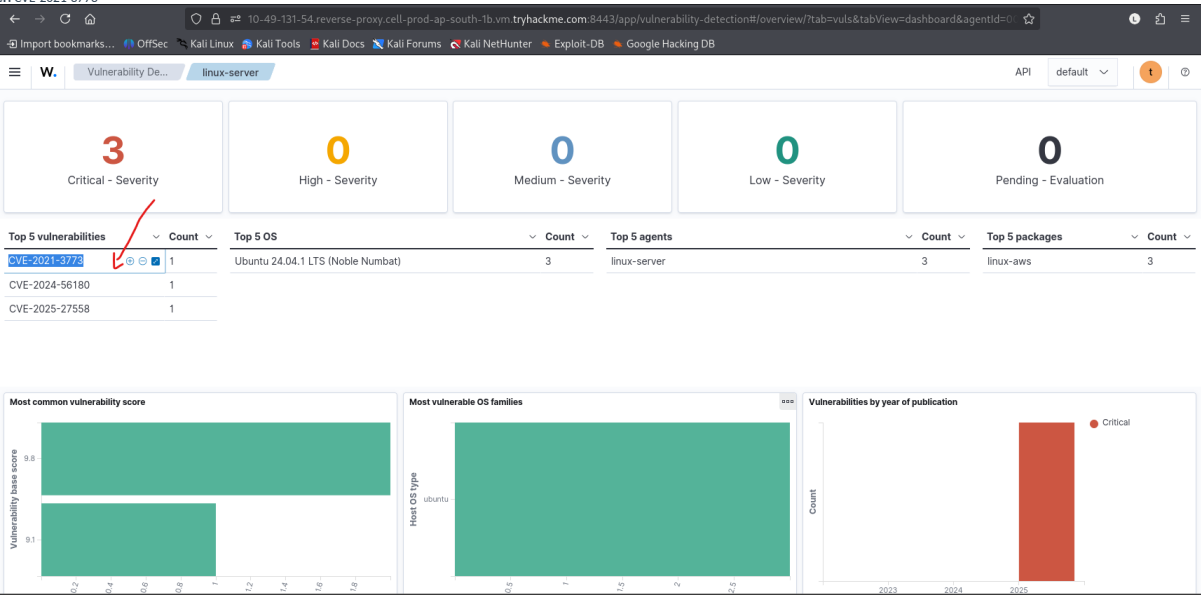
Table Headers: agent.name, package.name, package.version, vulnerability.description, vulnerability.severity, vulnerability.id

Table Data (selected rows):

agent.name	package.name	package.version	vulnerability.description	vulnerability.severity	vulnerability.id
WIN-SERVER	Notepad++ (64-bit x64)	8.4	Notepad++ versions prior to 8.8.9, wh...	High	CVE-2025-15556
WIN-SERVER	Notepad++ (64-bit x64)	8.4	Notepad++ is a free and open-source ...	Medium	CVE-2023-40164
WIN-SERVER	Notepad++ (64-bit x64)	8.4	Notepad++ is a free and open-source ...	Medium	CVE-2023-40166
WIN-SERVER	Notepad++ (64-bit x64)	8.4	Notepad++ v8.4.1 was discovered to ...	Medium	CVE-2022-31902
WIN-SERVER	Notepad++ (64-bit x64)	8.4	Notepad++ is a free and open-source ...	High	CVE-2023-40031
WIN-SERVER	Notepad++ (64-bit x64)	8.4	Buffer overflow in function Notepad_p...	Medium	CVE-2022-31901
WIN-SERVER	Notepad++ (64-bit x64)	8.4	Notepad++ versions 8.4.1 and before ...	High	CVE-2022-32168
WIN-SERVER	Notepad++ (64-bit x64)	8.4	Notepad++ is a free and open-source ...	High	CVE-2026-25926
WIN-SERVER	Notepad++ (64-bit x64)	8.4	Notepad++ is a free and open-source ...	Medium	CVE-2023-40036



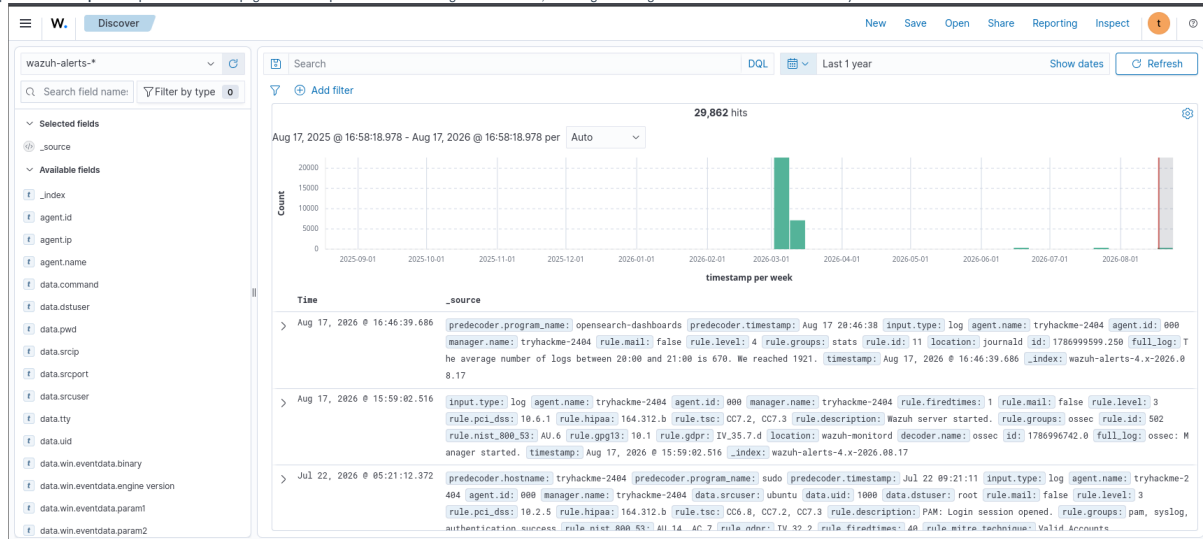
Question: What is the earliest critical vulnerability found on the Linux agent? Provide the oldest CVE number (e.g., CVE-2015-3719).
Answer: CVE-2021-3773



Task 6: Logging and Reporting

Logging with Wazuh

The core Wazuh capability is to collect and parse security logs from the agents. With some tweaks, Wazuh becomes a powerful SIEM, similar to Elasticsearch and Kibana. Let's start by checking the authentication logs from the Linux server. First, open the top-left menu > **Explore** dropdown > **Discover** page. This is the panel where all raw logs can be viewed, including the configuration assessment and vulnerability detection results:



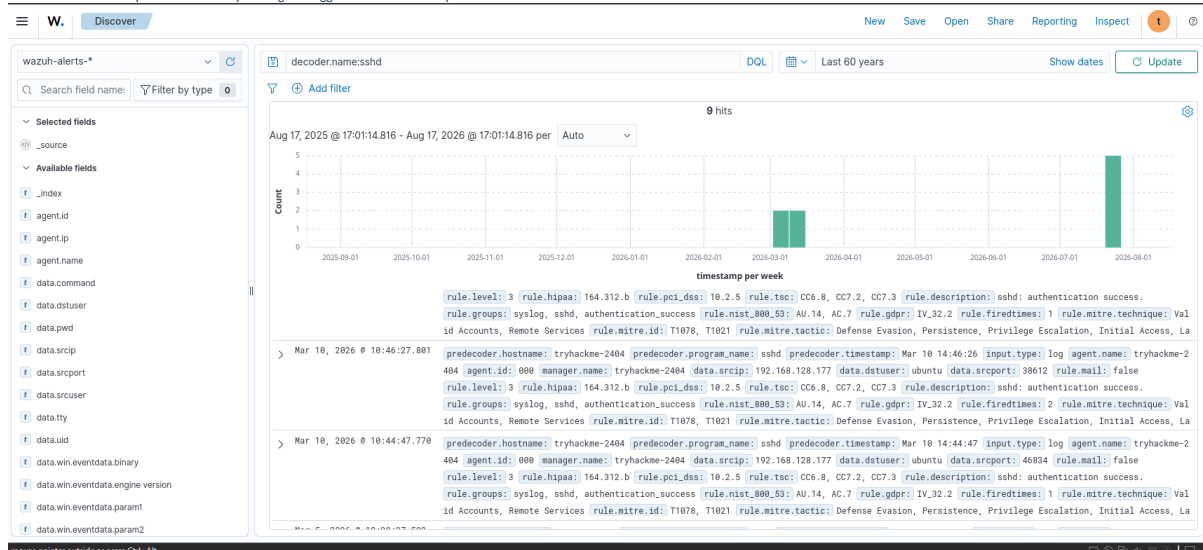
Next, let's explore the Discover menu:

Index pattern selector and field search: Useful to learn which fields are present in the events.

Search query input and time selector: To view logins, please apply these settings:

Search query: **decoder.name: sshd**

Time range: **Last 5 years**
The actual events: You can expand each event by clicking the toggle near the timestamp.

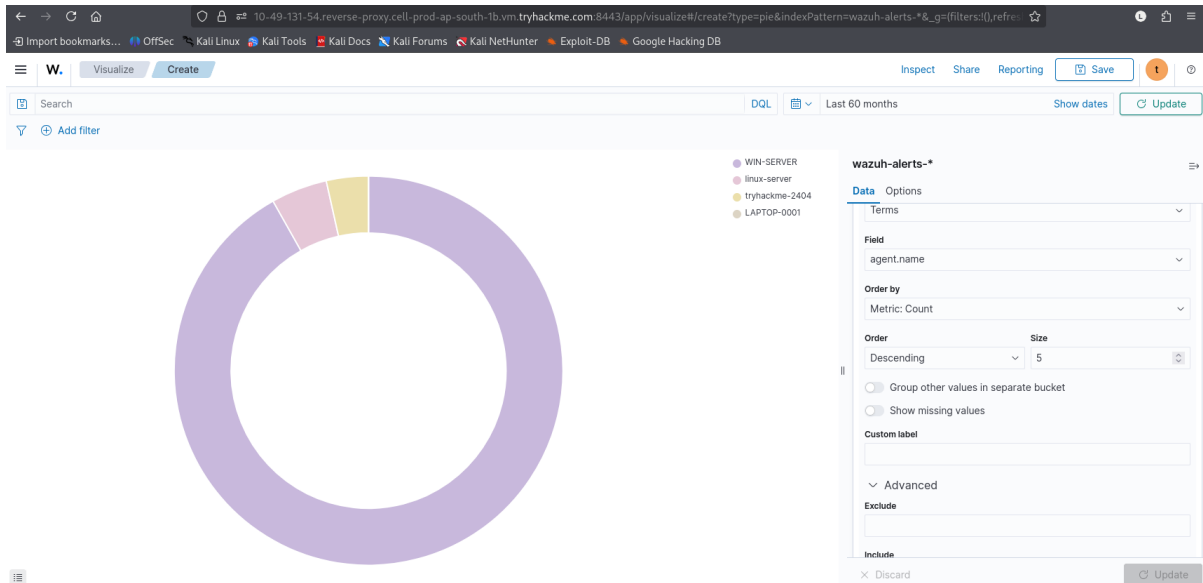


Logs vs. Rules vs. Alerts

In Wazuh, there is no concept of traditional SIEM logs. Instead, you define what logs will become alerts using Wazuh rules, and all non-matching events will be discarded. For example, the two SSH events you saw earlier were Wazuh alerts, generated by rule ID 5715. Wazuh has hundreds of prebuilt rules for different OS data sources, each having a level from 1 to 15, where 15 is the most critical alert possible. We will explore the rules in the next room.

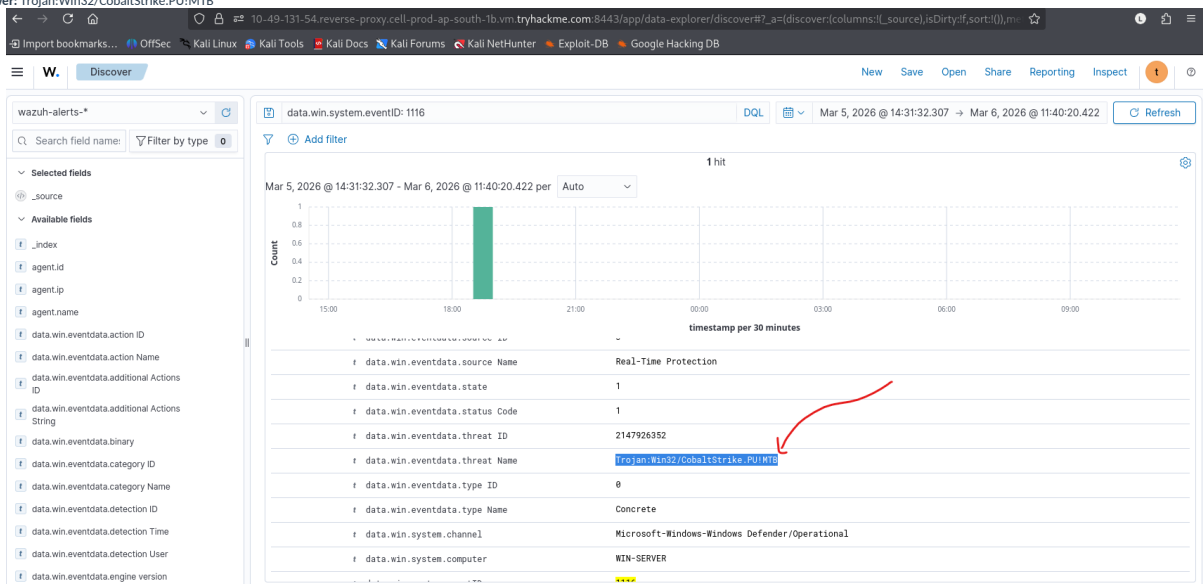
Dashboards and Reports

While Wazuh has tons of prebuilt dashboards for routine tasks, you might want to create a custom one at some point. To begin with, you can open the top-left menu > **Explore** > **Visualize** > **Create Visualization**. From there, select **Pie**, choose the **wazuh-alerts-*** source, add the **Split slices** bucket, and aggregate by **agent . name**, as shown in the image below. Then, click **Update**. Don't forget to set the global time range to **Last 5 years**.

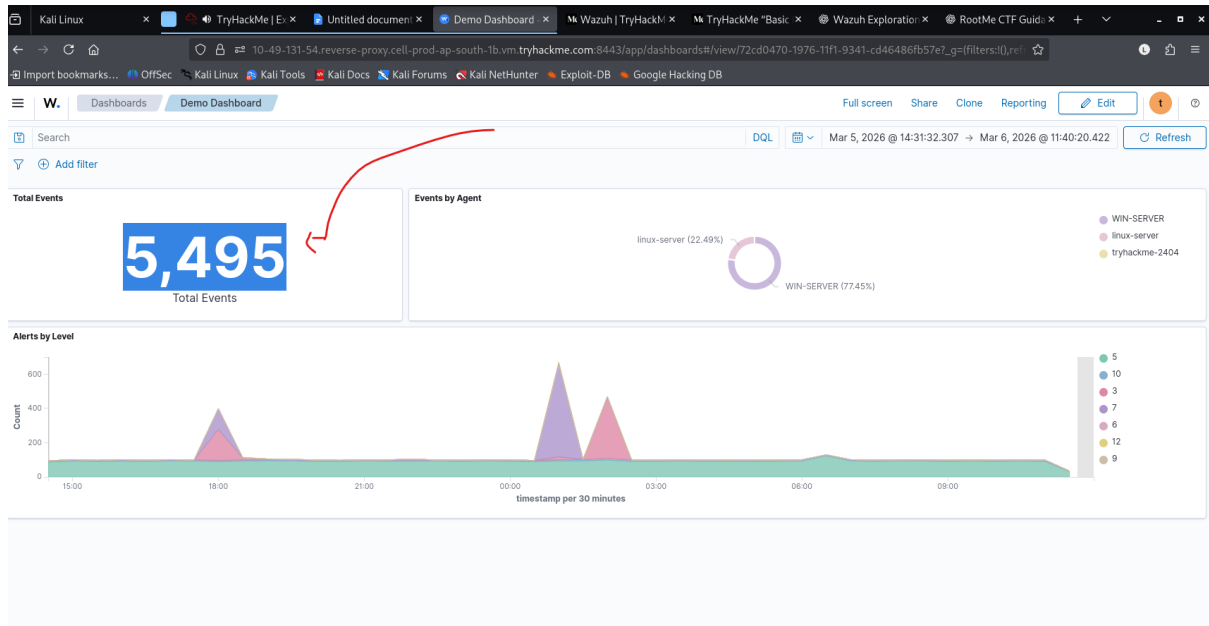


You can combine multiple visualization panels into a single dashboard. You can explore the prepared Demo Dashboard by navigating to the top-left menu > **Explore** > **Dashboards** menu. The menu might be overwhelming at first, but overall, Wazuh has very powerful reporting capabilities.

- **Question:** Navigate to Discover and search for Defender alerts for the last 5 years using the query `data.win.system.eventID: 1116`. What is the threat name of the detected malware?
- **Answer:** Trojan:Win32/CobaltStrike.PUI.MTB



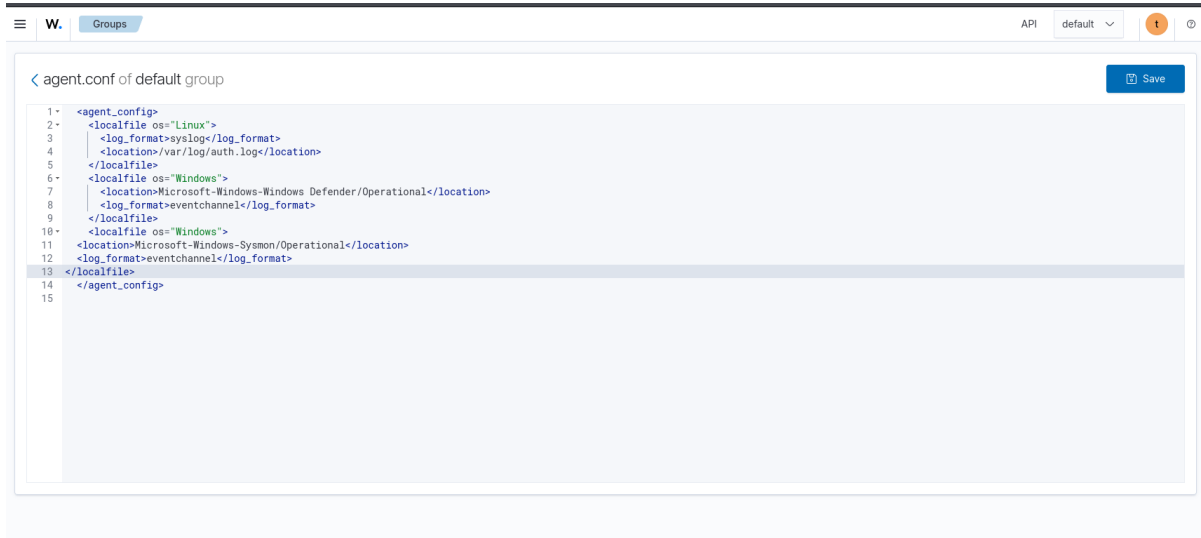
- **Question:** Navigate to the Demo Dashboard. How many total events are shown?
- **Answer:** 5,495



Task 7: Custom Rules and Log Collection

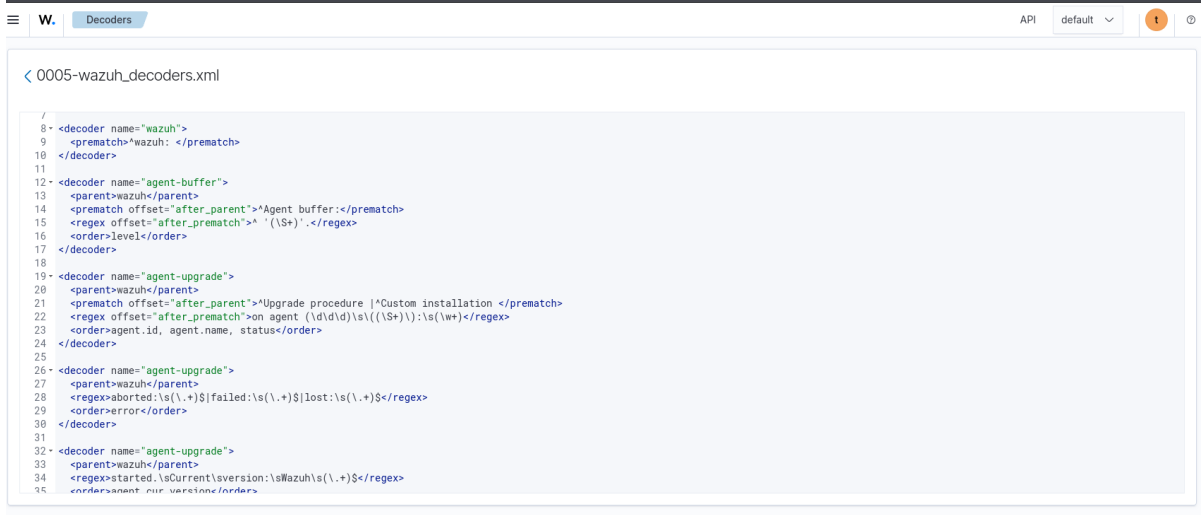
Agent Configuration

Wazuh collects basic system logs by default, but that's usually not enough for mature SOC teams. To collect custom logs, you need to specify the requirements in the **Groups** menu, and they will automatically propagate to all active agents. We have already added two custom settings to the default group to collect `/var/log/auth.log` from the Linux agent and Microsoft Defender event logs from Windows agents. Check it out by going to **Agents management > Groups** and clicking the pencil icon:



Wazuh Decoders

The agent configuration defines what to collect from the agents. Next, you need to instruct Wazuh on how to process (parse) the incoming raw data, as every source sends the events in its own unique format. The parsing instructions are called decoders, and there are hundreds of default decoders you can find in the **Server management > Decoders** menu. We won't explore the decoders in this room, but here is an example for Sysmon event code 1 for your reference:



Wazuh Rules

Next, the parsed logs must become alerts to be visible in the Wazuh dashboard. To create alerts, you need to create Wazuh rules. For example, in order to monitor PowerShell execution on your agents, you will need two rules: the first to alert on all Sysmon event code 1 logs (`level="0"` means informational severity), and the second to alert on any PowerShell execution (`level="12"` means a high-importance event). You can see the existing rules in the **Server management > Rules** menu.

W. Rules API default t

Rules (4,513)

From here you can manage your rules.

Search WQL Custom rules

ID ↑	Description	Groups	Level	File	Path
1	Generic template for all syslog rules.	syslog	0	0010-rules_config.xml	ruleset/rules
2	Generic template for all firewall rules.	firewall	0	0010-rules_config.xml	ruleset/rules
3	Generic template for all ids rules.	ids	0	0010-rules_config.xml	ruleset/rules
4	Generic template for all web rules.	web-log	0	0010-rules_config.xml	ruleset/rules
5	Generic template for all web proxy rules.	squid	0	0010-rules_config.xml	ruleset/rules
6	Generic template for all windows rules.	windows	0	0010-rules_config.xml	ruleset/rules
7	Generic template for all wazuh rules.	ossec	0	0010-rules_config.xml	ruleset/rules
200	Grouping of wazuh rules.	wazuh	0	0016-wazuh_rules.xml	ruleset/rules
201	Agent event queue rule	agent_flooding, wazuh	0	0016-wazuh_rules.xml	ruleset/rules
202	Agent event queue is level full.	agent_flooding, wazuh	7	0016-wazuh_rules.xml	ruleset/rules

Rows per page: 10 < 1 2 3 4 5 ... 452 >

Alert Notifications

Finally, you can configure notifications if some of your rules are triggered. In the simplest case, you can write a query `rule.level >= 12`, create an alert from it in **Explore > Alerting** menu, and then set up a notification channel (e.g., Slack message or email). You will receive a notification every time your query matches new events.

W. Notification cha... Notification ch... Channels Create channel t

Description - optional

✓ Slack

Chime

Microsoft Teams

Custom webhook

Email

Amazon SNS

Slack

Slack webhook URL

https://hooks.slack.com/services/XXXXX/XXXXX/XXXXX...

- Question: How do you call the Wazuh element that extracts fields from raw logs?
- Answer: Decoder
- Question: What Wazuh alert is more critical, the one with rule.level set to 10 or 15?
- Answer: 15

Task 9: Unique Wazuh Features

Explored Features

Let's summarize the features we have explored. For IT and GRC, these include:

- **Vulnerability Detection** to track software vulnerabilities on the agents.
- **Configuration Assessment** to track security misconfigurations on the agents.
- **IT Hygiene** to view agent details and list installed apps and browser extensions.

There are also several great features to make Wazuh a powerful **SIEM**, such as:

- **Decoders and Rules** to ingest and monitor system and application logs.
- **Discover and Dashboards** to search and visualize the ingested logs.

Now, let's explore some lesser-known Wazuh features that you should definitely test yourself!

Wazuh Active Response

In response to received events, Wazuh can send a response command ([Active Response](#)) to the agent: run a script, delete a file, block a network connection, and more. Just keep in mind that the response is executed with a delay (up to a few minutes), and therefore cannot fully replace dedicated endpoint security features like real-time process kill or malware quarantine. Still, Active Response can be used as an RMM/MDM for mass software deployment or to support the existing antivirus/EDR solution.

File Integrity Monitoring

Many government regulations require organizations to monitor critical files for unauthorized changes, including OS configurations, application settings, and user profiles. Beyond compliance, companies may also need to enforce strict access controls on sensitive files such as SSH keys. Wazuh addresses both needs through its [File Integrity Monitoring \(FIM\)](#) module, which logs every modification to selected files and directories (as well as registry keys on Windows). Explore it yourself under **Endpoint Security > File Integrity Monitoring**!

FIM and Malware Detection

Wazuh goes beyond basic file monitoring with two built-in malware detection features: YARA and VirusTotal integration. Whenever a monitored file is created or modified, Wazuh automatically scans it using your YARA rules and checks its hash against VirusTotal. A match triggers an alert and, optionally, an Active Response action to contain the threat. Note that you must bring your own YARA rules and VirusTotal key to make the features work.

Wazuh is a universal security solution that can fit different needs and be used by different teams. Try installing Wazuh in your home lab—it's highly educational! Also, if you want to play with Wazuh without ingesting real logs, try ingesting the demo data via **Indexer management > Sample Data**. Just note that some datasets may require an internet connection to be downloaded. Hope you enjoyed the room!